

The Value-Creation Bridge — Practitioner’s Guide

How to run the analysis: data, formulas, classification rules, worked end-to-end

Methods companion to the one-pager — for analysts (**internal**)

2 June 2026

What this is. The one-pager *sells* the Value-Creation Bridge; this guide *runs* it. It is the document you reach for when a client (or a skeptical colleague) says “show me how the numbers are actually built.” It gives the data, the exact formulas, the classification decision-rules, and a single worked example carried through end to end — the same deal as the one-pager, so the figures reconcile.

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1 The principle and the two splits

One rule governs everything: **the bridge must reconcile to the realized equity gain to the dollar**. Value is conserved — every dollar of gain is either value the business *created* or value *transferred* across a boundary, and within the created pool every dollar traces to one operating lever. Nothing is left in an “interaction” or “other” line. The analysis has two splits:

1. **Split 1 — Created vs. Transferred.** Decompose the equity gain into operational value created versus value transferred in (multiple re-rating, deleveraging, tax, timing).
2. **Split 2 — Attribution within Created.** Decompose the created pool across **pricing, volume, and cost productivity**, residual-free.

2 Step 0 — Define the system boundary

Before any arithmetic, name the counterparties, because “transfer” means *across this boundary*: the company, its **lenders**, the **tax authority**, the **prior and next owners**, and **customers**. Value that grows the whole system is created; value that only moves between these parties is transferred. *This choice is an input — state it.*

3 Split 1 — Created vs. Transferred

3.1 Data required

Entry and exit **EBITDA** (E_0, E_1), **EV/EBITDA multiple** (m_0, m_1), and **net debt** (D_0, D_1). Equity value is $\text{Equity} = E \cdot m - D$.

3.2 The bridge identity

The equity gain splits as

$$\Delta \text{Equity} = \underbrace{\Delta \text{EV}}_{E_1 m_1 - E_0 m_0} - \Delta D, \quad \Delta \text{EV} = \underbrace{\Delta E \cdot m_0}_{\text{EBITDA growth}} + \underbrace{\Delta m \cdot E_0}_{\text{re-rating}} + \underbrace{\Delta E \cdot \Delta m}_{\text{interaction}}.$$

The third term is the crux: a conventional bridge dumps it into a residual. **Do not.** Allocate it by a single, stated rule:

- **Symmetric (Shapley — canonical).** Split the interaction 50/50, which is identical to valuing *each driver at the average of the other*:

$$\text{EBITDA growth} = \Delta E \cdot \bar{m}, \quad \text{re-rating} = \Delta m \cdot \bar{E}, \quad \bar{m} = \frac{m_0 + m_1}{2}, \quad \bar{E} = \frac{E_0 + E_1}{2}.$$

This is the uniquely symmetric, residual-free allocation — the rule the methodology rests on.

- **Sequential (simpler).** Grow EBITDA at the entry multiple, then re-rate at exit EBITDA: $\Delta E \cdot m_0$ and $\Delta m \cdot E_1$. Folds the interaction into re-rating.

Pick one, state it, and report both if the difference is material.

3.3 Classification rules (created vs. transferred)

- **Created — EBITDA growth:** real operating improvement, valued at a *constant* multiple so growth is never conflated with re-rating.

- **Transferred — Multiple re-rating:** the market/next buyer paying a different multiple. *Default: transfer.*
- **Transferred — Deleveraging:** debt paydown shifting enterprise value from the debt claim to the equity claim. *Default: transfer* (enterprise value is unchanged by the paydown itself).
- **Transferred — Tax shields, timing/market beta:** value from the state or the cycle.

Judgment calls — decide and document. (1) If a re-rating reflects genuine structural de-risking the company achieved (recurring revenue, lower customer concentration), a portion of multiple expansion may be reclassified as **Created** — with justification. (2) The cash flow that *funded* deleveraging is operational; a sponsor who credits that cash conversion to operations reclassifies the funded portion as **Created**. State the choice; the bridge updates.

3.4 Worked example

$E_0=100$, $E_1=150$; $m_0=10.0$, $m_1=11.0$; $D_0=600$, $D_1=400$. Then $\text{Equity}_0=400$, $\text{Equity}_1=1,250$, gain = \$850M. Using the sequential convention (matching the one-pager):

Source	\$M	Formula	Bucket
EBITDA growth	500	$\Delta E \cdot m_0 = 50 \times 10$	Created
Multiple re-rating	150	$\Delta m \cdot E_1 = 1 \times 150$	Transferred
Deleveraging	200	$-\Delta D = 600 - 400$	Transferred
Equity gain	850		

Created \$500M (59%); transferred \$350M (41%). The “created MOIC”: value exit EBITDA at the *entry* multiple on the *entry* balance sheet, $E_1 m_0 - D_0 = 1,500 - 600 = \900M , so $900/400 = 2.25\times$ — and $900 - 400 = \$500\text{M}$, the created pool. The headline $3.1\times$ is a $2.25\times$ of created value wrapped in financial engineering.

Convention check (robustness). The symmetric/Shapley rule gives EBITDA growth $\Delta E \cdot \bar{m} = 50 \times 10.5 = \525M and re-rating $\Delta m \cdot \bar{E} = 1 \times 125 = \125M — so created = \$525M (62%). The two conventions differ by \$25M (~ 3 points); the conclusion — “well under two-thirds was created” — is robust to the choice. Report the convention you used.

4 Split 2 — Attribution within Created

4.1 Data required — and the deflator step

EBITDA factors as $E = P \cdot Q \cdot \mu$ — **price index** $\times$ **volume index** $\times$ **margin**. Revenue = $P \cdot Q$; margin $\mu = E/\text{Revenue}$. To separate price from volume you need a **realized net price index** P (list price changes net of discounts, revenue-weighted across lines); then volume $Q = \text{Revenue}/P$. *The deflator is an input — state its source.*

4.2 The decomposition (log / Divisia — residual-free)

Because $E = P \cdot Q \cdot \mu$ is a product, logs add exactly:

$$\Delta \ln E = \Delta \ln P + \Delta \ln Q + \Delta \ln \mu.$$

Each lever's share of the created value is its share of log-growth:

$$\text{contribution}_i = \frac{\Delta \ln(\text{factor}_i)}{\Delta \ln E} \times (\text{created value}).$$

There is no cross-term: the shares sum to 1 by construction, so the dollars sum to the created pool with no residual.

4.3 The Aumann–Shapley alternative (levels, and ROIC)

When the structure is additive or you decompose a *level* (e.g. ROIC = margin $\times$ turns), use the symmetric line-integral: split each pairwise interaction 50/50 (a triple interaction $\frac{1}{3}$ each). For ROIC: $\Delta \text{ROIC} = \Delta(\text{margin}) \cdot \overline{\text{turns}} + \Delta(\text{turns}) \cdot \overline{\text{margin}}$. This is the same “value each factor at the average of the others” rule as Split 1, and it is path-independent — the unique residual-free level attribution.

4.4 Worked example

Carrying the created \$500M. Revenue grew $500 \rightarrow 641$ (+28.2%) and margin $20\% \rightarrow 23.4\%$ (+17.0%), so EBITDA $100 \rightarrow 150$ ($\times 1.50$). The factors compound exactly: $1.080 \times 1.187 \times 1.170 = 1.50$.

Lever	Factor	$\Delta \ln$ / share	\$M
Pricing (+8.0%)	1.080	0.0770 / 19%	95
Volume (+18.7%)	1.187	0.1714 / 42%	211
Cost productivity (+17.0%)	1.170	0.1570 / 39%	194
Total	1.500	0.4054 / 100%	500

Read: volume and cost productivity did $\sim 80\%$ of the work; pricing only 19% — the under-exploited lever for the next hold.

5 Reconciliation — always close the loop

The final check, every time: the buckets must sum back to the realized equity gain.

$$\underbrace{(95 + 211 + 194)}_{\text{created 500}} + \underbrace{150}_{\text{re-rating}} + \underbrace{200}_{\text{deleveraging}} = \$850\text{M} = \Delta \text{Equity}. \checkmark$$

If it does not tie to the dollar, a flow has been mis-mapped or double-counted — find it before presenting.

6 Judgment checkpoints (the inputs the math cannot supply)

The arithmetic is exact; the *answer* is only as good as four inputs, each of which must be stated as an assumption:

1. **System boundary** — who counts as a counterparty (Step 0).
2. **Classification** — the created-vs-transferred calls, especially re-rating (de-risking?) and deleveraging (credit the cash conversion?).
3. **Interaction convention** — symmetric (canonical) or sequential.
4. **Deflator** — the price index separating pricing from volume.

Change any one and the bridge updates transparently. That is the methodology's discipline: the assumptions are visible and contestable, not buried.

7 Why the decomposition works (self-contained)

You don't have to take the exactness on faith — each piece has an elementary reason, and none of it reaches beyond linear algebra, logs, and a path integral.

- **Created + transferred = the whole, no overlap.** Write the per-stakeholder value change as a vector d . Split it into its average component $\bar{d}\mathbf{1}$ (the same for everyone — this carries the change in the *total*, so $\sum \text{created} = \sum d$) and the remainder $d - \bar{d}\mathbf{1}$ (which sums to zero — pure redistribution). The two are orthogonal: $\langle \mathbf{1}, d - \bar{d}\mathbf{1} \rangle = \sum_i (d_i - \bar{d}) = 0$. Orthogonal pieces obey Pythagoras, so $\|d\|^2 = \|\text{created}\|^2 + \|\text{transferred}\|^2$ — there is *no cross-term*. (It is the projection of d onto the all-ones line and its complement.)
- **Why the lever split has no residual.** $\text{EBITDA} = \text{price} \times \text{volume} \times \text{margin}$, so $\ln \text{EBITDA} = \ln p + \ln v + \ln m$ and the changes add *exactly*: $\Delta \ln \text{EBITDA} = \Delta \ln p + \Delta \ln v + \Delta \ln m$. The log of a product is a sum, so no interaction term can survive — the shares are exhaustive by construction.
- **Why the symmetric split is exact *and* canonical.** For a product, $\Delta(xy) = \Delta x y_0 + x_0 \Delta y + \Delta x \Delta y$; splitting the interaction $\Delta x \Delta y$ in half gives each factor its change valued at the *average* of the other, and the two add back to $\Delta(xy)$ exactly (expand and check). It is canonical because it is the integral of the gradient along the straight path from start to finish — the only allocation that treats the factors identically (neither goes “first”). For three factors the same path integral produces the $\frac{1}{2} / \frac{1}{3}$ weights, since $\int_0^1 s ds = \frac{1}{2}$ and $\int_0^1 s^2 ds = \frac{1}{3}$; and a path integral from a to b equals $f(b) - f(a)$, which is exactly why nothing is left over.
- **Why deleveraging is a transfer.** Equity = EV – net debt; paying down debt with the firm's cash leaves EV unchanged (you retire a liability with an asset), so the equity gain is value moving from the debt claim to the equity claim within a fixed enterprise value — a transfer between claimholders, not new value. (The cash that funded it *was* operational — the judgment call flagged in the assumptions.)

What's exact is the arithmetic structure — the parts sum to the whole with no double-counting. **The four inputs** — boundary, classification, interaction convention, deflator — **remain judgment**. Present it as “exact and transparent: check it on the page,” never as an external proof.

Methods companion to “The Value-Creation Bridge.” Draft prepared by the ENT 105 team for Creo Advisors. Figures are illustrative and reconcile with the one-pager. The decomposition mechanics are grounded in conservation-based

accounting; the boundary, classification, interaction convention, and deflator are client-specific inputs set at the engagement's scoping stage.